RS485 typically only uses A and B. Unless shielded cables are used.
DMX typically uses A, B and GND (Shield).
RS485 Ground is Isolated from Board Ground!

[UART]

Galvanically Isolated

RS485 | ESD | Surge | MODBUS | DMX512

* 120R Load/Terminating End-of-Line Resistor. Default in Circuit.

Max 30V

ADC2/3 2 x Voltage Ranges

Inputs

I/O I2C | OLED

Analog ADC

VDC Voltage Monitor

16bit ADC

Absolute Max -0.3V to 3.6V

Address 0x49

VDC Approx
05V - 0.17V
12V - 0.46V
20V - 0.78V
30V - 1.18V
40V - 1.57V
50V - 1.97V
60V - 2.36V
75V - 2.96V
80V - 3.16V

Apprx. mA/V
04 - 0.64
10 - 1.60
15 - 2.40
20 - 3.20

4-20mA

Opto Galvanic Insulated Inputs

Max 75V With Jumper

ADC Inputs | LV or HV

Temperature | TMP102

Address 0x4A

Expansion

U.FL Version uses IPEX MHF3 Coax Connectors

~Mandatory Boot States
GP8 High
GP9 High

Variants:
* ESP32-C6-MINI-1U-H4

- Bluetooth
  Bluetooth LE
  Bluetooth 5.3
  Bluetooth Mesh
- Wi-Fi 6
  2.4 GHz
  802.11ax (20 MHz bandwidth)
  802.11b/g/n (20/40 MHz BW)
- Thread 1.4
  802.15.4
  Thread Mesh
- Zigbee 3.0
  802.15.4
  Zigbee Mesh

MCU | ESP32-C6 | EEPROM | 24C64

Address 0x50

Typical Current Approx 12mA @ 5V

Status | RGB | LED

User LED and Heatbeat Use

User Button
* Force PGM
* * User

Tactile Switches | Reset | Usr/Pgm

Thermal, Short and Over Current Protection
3V3 powers all devices (except RGB LED).

700mA Max

5V (VBUS) to 3V3 LDO PSU

[SPI]

SX1262 is compatible with Radio Tranceiver SX1278 and SX1276

868 MHz
915 MHz

LoRa | EByte E22-900MM22S | SX1262

NO
Common
NC

SRD-05VDC

Relay Isolated

Default States P:

Inputs
0-H
1-H
2-L
3-H

Outputs
4-F
5-F
6-F
7-F

Dual Reset

GPIO State Resets on Power Cycle

Address 0x21

FET Sink 230mA 60V Max

OPTO Sink 50mA 35V Max

GPIO Matrix | Outputs

CR2032 3V

Address 0x68

DS3231 | RTC | VBatt

USB UART providing Arduino/PlatformIO Flashing ESP32-C6 DEV MODULE Compatibility
Board compatible with esp32-c6-devkitm-1 and adafruit_feather_esp32c6

GPIO9 + Reset

Flash Boot	High
UART Download	Low

CC1 and CC2, are used to establish and manage the Source-to-Sink connection

USB 1 | USB UART | Serial Monitoring | USB Flashing | Debug | ESD

Auto Flash

High Voltage Power Suppy | HVDC SMPS 5V DC-DC | ADC

External DC Input DC In/Out

! Beware of any High Voltages

DC In 5-60V. Maximum 75V Input.
Permanant damage WILL occur > 75V !
* Reverse Polarity Protected
* Fuse Hold 500mA Trip 1A Resettable

* DC In Sense Voltage to ADC0 (VDC) Input

DC Input LED

600mA Max

SMPS 24V

250 mA Max

SPI EXT

Enclosure VG-GR35 90x84x57 Main PCB 86x79.5

For further information and code examples, please see our code and GitHub WIKI pages. ditronix.net | GitHub.com/DitroniX | Hackster.io/DitroniX

Components optimised for BOM

DitroniX
Home Automation | Internet of Things | Energy Monitoring

Production

(C)2021-2026 - DitroniX - Dave Williams G8PUO

ESP32-C6 SX1262 SDK **ditronix.net** DitroniX Ltd – UK	ESPKnack ESP32-C6 SX1262 SDK 1.2604.105		
	ESPKNACK-LORA-ESP32-C6-GALV-ISOLATED-CONTROLLER-SDK		
	Sheet: 1/1	Drwg/Chk: **Dave Williams**	28/04/2026 16:47